Rule-State Inference (RSI) — Supplementary Material

Appendices A, B, C

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This document contains supplementary material for the paper *Rule-State Inference (RSI): A Bayesian Framework for Compliance Monitoring in Rule-Governed Domains*. It includes the dataset dictionary (Appendix A), complete theorem proofs (Appendix B), and reproducibility details (Appendix C).

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A Dataset Dictionary: RSI-Togo-Fiscal-Synthetic v1.0

A.1 Overview

The dataset contains 2,000 synthetic enterprises across two regulatory periods (2022–2024 and 2025), generated from real OTR fiscal rules. Each row is structured in four layers:

1. **Enterprise features**: observable entity attributes
2. **Latent ground truth** (`gt_*`): the true rule state (a_i, c_i, δ_i), never seen by RSI during inference
3. **Noisy observations** (`obs_*`): what a fiscal system actually observes
4. **Binary labels** (`label_*`): derived compliance labels for baseline evaluation

A.2 Column Reference

Column	Type	Description
Identifiers		
`enterprise_id`	string	Unique enterprise identifier (e.g., ENT_00001)
`period`	string	Regulatory period: `2022_2024` or `2025`
Enterprise features		
`sector`	string	Economic sector (commerce, services, BTP, industrie, agriculture, restauration, transport)
`region`	string	Geographic region in Togo (Lomé, Sokodé, Kara, etc.)
`ca_segment`	string	Turnover segment: `informal` (<30M), `rsi` (30–100M), `reel` (≥100M)
`n_employes`	int	True number of employees (log-normal distribution)
Latent ground truth (gt_*) — not available to RSI during inference		
`gt_ca_reel`	float	True annual turnover in FCFA
`gt_benefice_reel`	float	True annual profit in FCFA
`gt_tva_active`	int	1 if VAT rule applies (CA $\geq$ threshold), 0 otherwise
`gt_tva_compliance`	float	True VAT compliance rate $c_i \in [0, 1]$, drawn from Beta prior
`gt_tva_due`	float	Theoretical VAT due (FCFA)
`gt_is_active`	int	1 if CIT rule applies (CA $\geq$ 100M FCFA)
`gt_is_compliance`	float	True CIT compliance rate $c_i \in [0, 1]$
`gt_is_due`	float	Effective CIT due (max of CIT and IMF) in FCFA
`gt_tpu_active`	int	1 if TPU rule applies (CA $<$ 30M FCFA)
`gt_tpu_compliance`	float	True TPU compliance rate $c_i \in [0, 1]$
`gt_irpp_susp_active`	int	1 if IRPP suspension applies (salary $<$ 900k, period 2023)
Noisy observations (obs_*) — RSI input		
`obs_ca_declare`	float	Declared turnover. Generated as $\hat{x} = x \cdot \beta \cdot \varepsilon$ where $\beta \sim \text{Beta}(7, 3)$ captures systematic under-declaration and $\varepsilon \sim \mathcal{N}(1, 0.045)$ captures measurement noise. Mean ratio $\mathbb{E}[\hat{x}/x] \approx 0.70$.
`obs_tva_declaree`	float	Declared VAT amount. Zero if entity does not declare; log-normally distributed around theoretical VAT if declared. Set to `NaN` with probability 0.18 (missing).
`obs_tva_assujetti_declare`	bool	Whether the entity self-declared as VAT-registered

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Column	Type	Description
`obs_is_declare`	float	Declared corporate income tax. Set to `NaN` with probability 0.18.
`obs_benefice_declare`	float	Declared profit. Typically 70% of true profit with noise. Set to `NaN` with probability 0.18.
`obs_retard_paiement_jours`	int	Payment delay in days. Generated from Exponential(λ) where $\lambda = 5/(c_{\text{regime}} + 0.1)$, with c_{regime} being the compliance of the applicable regime. Higher non-compliance $\Rightarrow$ longer delays.
`obs_has_compte_bancaire`	bool	Whether entity has a formal bank account. $P(\text{account}) = 0.75 \cdot c + 0.20 \cdot (1 - c)$ where c is regime compliance.
`obs_utilise_facturation_electronique`	bool	Whether entity uses electronic invoicing
`obs_a_ete_audite`	bool	Whether entity was audited in the period (8% base rate)
`obs_n_employes_declare`	int	Declared number of employees ($\approx$ 85% of true value)
`obs_ratio_sous_declaration`	float	Ratio of declared to true turnover ($\hat{x}/x$)
`obs_tva_missing`	bool	True if `obs_tva_declaree` is missing
`obs_is_missing`	bool	True if `obs_is_declare` is missing
`obs_benefice_missing`	bool	True if `obs_benefice_declare` is missing
Binary labels (label_*) — used only for baseline evaluation		
`label_tva_non_conforme`	int	1 if VAT active and $c_{\text{TVA}} < 0.5$
`label_is_non_conforme`	int	1 if CIT active and $c_{\text{IS}} < 0.5$
`label_any_non_conforme`	int	1 if any of: VAT non-compliant, CIT non-compliant, or TPU compliance < 0.3

A.3 Under-declaration Model

The under-declaration ratio is the key noise mechanism reflecting real-world fiscal behavior in Francophone Africa. For each enterprise:

$$\hat{x} = x \cdot \underbrace{\beta}_{\text{systematic}} \cdot \underbrace{\varepsilon}_{\text{noise}}, \quad \beta \sim \text{Beta}(7, 3), \quad \varepsilon \sim \mathcal{N}(1,\, 0.045^2)$$

The Beta(7,3) distribution has mean $7/10 = 0.70$ and concentrates mass between 0.55 and 0.85, consistent with empirical estimates from Sub-Saharan African fiscal studies. The Gaussian noise term captures accounting approximations and rounding.

A.4 Regulatory Change Event

The period `2025` encodes the VAT threshold change from 60M to 100M FCFA enacted by Law n°2024-007 (30 December 2024). Enterprises with true turnover in $[60\text{M}, 100\text{M})$ FCFA shift from `gt_tva_active=1` (period 2022–2024) to `gt_tva_active=0` (period 2025), providing a clean natural experiment for evaluating Theorem 1 (O(1) adaptability).

A.5 Power-Law Structure

Turnover is drawn from a log-normal distribution within each segment, whose aggregate approximates a power law — the universal signature of firm size distributions in market economies [Gabaix(2009)]. The three-segment structure reproduces the concentration typical of African markets:

Segment	CA range	Share
Informal (TPU)	$<$ 30M FCFA	60%
Intermediate (RSI)	30M – 100M FCFA	25%
Large (CIT/VAT)	$\geq$ 100M FCFA	15%

B Complete Theorem Proofs

B.1 Proof of Theorem 1: O(1) Regulatory Adaptability

We provide the complete proof including the numerical stability argument.

Theorem 1 (O(1) Adaptability — complete)**.** *For any regulatory update $\mathcal{U}_k : \Theta_k \to \Theta'_k$, the RSI update cost is $\mathcal{C}^{RSI}(\mathcal{U}_k) = O(1)$, independent of $|\mathbf{D}|$ and n. Moreover, the updated posterior is properly normalized with no risk of variance explosion.*

Proof. **Step 1: Factorization.** The RSI prior factorizes as:

$$P(\mathbf{S}) = \prod_{i=1}^{n} P(s_i \mid \Theta_i)$$

After update $\mathcal{U}_k$, the new prior is:

$$P'(\mathbf{S}) = P(s'_k \mid \Theta'_k) \cdot \prod_{i \neq k} P(s_i \mid \Theta_i)$$

Step 2: Posterior correction. By Bayes' theorem, the posterior before the update is:

$$P(\mathbf{S} \mid \mathbf{D}) = \frac{P(\mathbf{D} \mid \mathbf{S}) \cdot P(\mathbf{S})}{P(\mathbf{D})}$$

After the update:

$$\begin{aligned} P'(\mathbf{S} \mid \mathbf{D}) &= \frac{P(\mathbf{D} \mid \mathbf{S}) \cdot P'(\mathbf{S})}{P'(\mathbf{D})} \\ &= \frac{P(\mathbf{D} \mid \mathbf{S}) \cdot P(s'_k \mid \Theta'_k) \cdot \prod_{i \neq k} P(s_i \mid \Theta_i)}{P'(\mathbf{D})} \\ &\propto P(\mathbf{S} \mid \mathbf{D}) \cdot \underbrace{\frac{P(s'_k \mid \Theta'_k)}{P(s_k \mid \Theta_k)}}_{\rho_k(\Theta_k, \Theta'_k)} \end{aligned}$$

Step 3: Complexity. The correction factor ρ_k depends only on the parametric forms of $P(s_k \mid \Theta_k)$ and $P(s'_k \mid \Theta'_k)$. Evaluating two parametric distributions (Bernoulli, Beta, Gaussian) requires a constant number of arithmetic operations, hence $O(1)$.

Step 4: Numerical stability. The ratio ρ_k is applied to the *kernel* (unnormalized form) of the posterior, not to its absolute value. The normalization constant is updated as:

$$P'(\mathbf{D}) = P(\mathbf{D}) \cdot \int_{\mathcal{S}} \rho_k(\Theta_k, \Theta'_k)\, dP(\mathbf{S} \mid \mathbf{D}) = P(\mathbf{D}) \cdot \mathbb{E}_{P(\mathbf{S}|\mathbf{D})}[\rho_k]$$

Under the conjugate families used in RSI (Beta-Binomial for c_i, Bernoulli for a_i, Gaussian-Gaussian for δ_i), this expectation is available in closed form. The posterior therefore remains a properly normalized probability distribution after the update, with no variance explosion. □

Corollary 1.1 (Cumulative cost over T updates)**.** *Over T regulatory updates, total costs scale as:*

Method	*Total cost*
RSI	$O(T)$
Supervised ML (full retrain)	$O(T \cdot \lvert\mathbf{D}\rvert \cdot E)$
PSL / MLN	$O(T \cdot \lvert\mathbf{D}\rvert \cdot n)$

where E denotes the number of training epochs. RSI dominates in environments with high regulatory change frequency.

B.2 Proof of Theorem 2: Bernstein-von Mises Consistency

We first state and verify the regularity conditions, then give the complete proof.

Regularity conditions.

C1 *Identifiability.* $P(\mathbf{D} \mid \mathbf{S}) \neq P(\mathbf{D} \mid \mathbf{S}')$ for $\mathbf{S} \neq \mathbf{S}'$. *Verification:* Each component (a_i, c_i, δ_i) influences the likelihood through distinct mechanisms (activation gating, Beta-distributed compliance, Gaussian drift). Distinct states produce distinct likelihood functions.

C2 *Positive prior at truth.* $P(\mathbf{S}^*) > 0$. *Verification:* The prior is a product of Bernoulli, Beta, and Gaussian distributions, all of which have strictly positive density on their respective supports. Hence $P(\mathbf{S}) > 0$ for all $\mathbf{S} \in \mathcal{S}$.

C3 *Regularity of the likelihood.* $\log P(\mathbf{D} \mid \mathbf{S})$ is twice differentiable in the continuous components (c_i, δ_i). *Verification:* The likelihood functions (log-Gaussian, exponential, Beta-Binomial) are smooth in their parameters. Differentiability is inherited by the product.

C4 *Finite Fisher information.* $\mathcal{I}(\mathbf{S}^*)$ is finite and positive definite. *Verification:* Under C1 and C3, and provided no rule is degenerate (i.e., $c_i \notin \{0, 1\}$ and $a_i \in (0, 1)$), the Fisher information matrix is positive definite.

Theorem 2 (BvM Consistency — complete)**.** *Under conditions C1–C4, as $m \to \infty$:*

$$\sqrt{m}\big(\hat{\mathbf{S}}_m - \mathbf{S}^*\big) \xrightarrow{d} \mathcal{N}\big(0, \mathcal{I}(\mathbf{S}^*)^{-1}\big)$$

Proof. **Step 1: LLN convergence.** By the Law of Large Numbers, for i.i.d. observations:

$$\frac{1}{m} \log P(\mathbf{D}_m \mid \mathbf{S}) \xrightarrow{a.s.} \mathbb{E}_{\mathbf{S}^*}[\log P(d \mid \mathbf{S})] = -\mathrm{KL}(P(\cdot \mid \mathbf{S}^*) \| P(\cdot \mid \mathbf{S})) + \text{const}$$

This is maximized at $\mathbf{S} = \mathbf{S}^*$ under C1.

Step 2: Local Gaussian approximation. By a second-order Taylor expansion of $\log P(\mathbf{D}_m \mid \mathbf{S})$ around $\mathbf{S}^*$:

$$\log P(\mathbf{D}_m \mid \mathbf{S}) \approx \log P(\mathbf{D}_m \mid \mathbf{S}^*) + \nabla \ell_m^\top (\mathbf{S} - \mathbf{S}^*) - \frac{1}{2}(\mathbf{S} - \mathbf{S}^*)^\top \big[-\nabla^2 \ell_m\big](\mathbf{S} - \mathbf{S}^*)$$

where $\ell_m = \log P(\mathbf{D}_m \mid \mathbf{S})$. By the LLN, $-\frac{1}{m}\nabla^2 \ell_m \xrightarrow{a.s.} \mathcal{I}(\mathbf{S}^*)$ (C3, C4).

Step 3: Posterior Gaussianization. Substituting into the posterior:

$$\begin{aligned} P(\mathbf{S} \mid \mathbf{D}_m) &\propto \exp(\log P(\mathbf{D}_m \mid \mathbf{S})) \cdot P(\mathbf{S}) \\ &\approx \exp\Big(-\frac{m}{2}(\mathbf{S} - \mathbf{S}^*)^\top \mathcal{I}(\mathbf{S}^*)(\mathbf{S} - \mathbf{S}^*)\Big) \cdot P(\mathbf{S}) \end{aligned}$$

Step 4: Prior dominance. Under C2, $P(\mathbf{S}) > 0$ in a neighborhood of $\mathbf{S}^*$. As $m \to \infty$, the exponential term concentrates mass at $\mathbf{S}^*$ at rate m, dominating the prior. The posterior converges to:

$$P(\mathbf{S} \mid \mathbf{D}_m) \xrightarrow{d} \mathcal{N}\left(\mathbf{S}^*,\ \frac{1}{m}\mathcal{I}(\mathbf{S}^*)^{-1}\right)$$

The stated result follows by the change of variables $\sqrt{m}(\mathbf{S} - \mathbf{S}^*)$. □

Corollary 2.1 (Prior robustness — complete)**.** *Let $P_1(\mathbf{S})$ and $P_2(\mathbf{S})$ be two RSI priors satisfying C2. Then:*

$$\mathrm{TV}\big(P_1(\mathbf{S} \mid \mathbf{D}_m),\ P_2(\mathbf{S} \mid \mathbf{D}_m)\big) \to 0 \quad \textit{as } m \to \infty$$

Proof. Both posteriors converge to $\mathcal{N}(\mathbf{S}^*, m^{-1}\mathcal{I}(\mathbf{S}^*)^{-1})$ by Theorem 2. Since total variation metrizes weak convergence on $\mathbb{R}^d$, $\mathrm{TV}(P_i(\mathbf{S}|\mathbf{D}_m), \mathcal{N}(\cdot)) \to 0$ for $i = 1, 2$, and the result follows by the triangle inequality. □

B.3 Proof of Theorem 3: Monotone ELBO Convergence

Theorem 3 (Monotone ELBO — complete)**.** *The RSI coordinate ascent updates produce a monotonically non-decreasing ELBO sequence:* $\mathrm{ELBO}_{t+1} \geq \mathrm{ELBO}_t$ *for all* $t \geq 0$.

Proof. Recall the ELBO decomposition:

$$\mathrm{ELBO}(\phi) = \mathbb{E}_{Q_\phi}[\log P(\mathbf{D} \mid \mathbf{S})] - \mathrm{KL}[Q_\phi(\mathbf{S}) \| P(\mathbf{S})] = \log P(\mathbf{D}) - \mathrm{KL}[Q_\phi(\mathbf{S}) \| P(\mathbf{S} \mid \mathbf{D})]$$

Since $\log P(\mathbf{D})$ is constant with respect to ϕ, maximizing the ELBO is equivalent to minimizing $\mathrm{KL}[Q_\phi \| P(\cdot|\mathbf{D})]$.

At iteration t, the update for factor i solves:

$$\phi_i^{(t+1)} = \arg\min_{\phi_i} \mathrm{KL}\big[Q_{\phi_i}(s_i) \cdot \prod_{j \neq i} Q_{\phi_j^{(t)}}(s_j) \parallel P(\mathbf{S} \mid \mathbf{D})\big]$$

The optimal solution is:

$$Q^*_{\phi_i}(s_i) \propto \exp\big(\mathbb{E}_{Q_{-i}}\big[\log P(\mathbf{D}, \mathbf{S})\big]\big)$$

This update can only decrease the KL divergence (or leave it unchanged), hence $\mathrm{ELBO}_{t+1} \geq \mathrm{ELBO}_t$.

Since the ELBO is bounded above by $\log P(\mathbf{D}) < \infty$, the sequence converges. The limit is a stationary point of the ELBO. □

Lemma 1 (ELBO gap bound)**.** *Let* Q^*_ϕ *denote the mean-field optimum. Then:*

$$\mathrm{KL}[Q^*_\phi(\mathbf{S}) \| P(\mathbf{S} \mid \mathbf{D})] \leq \sum_{i=1}^{n} \mathrm{KL}[Q^*_{\phi_i}(s_i) \| P(s_i \mid \mathbf{D})]$$

Proof. By the sub-additivity of KL divergence under the mean-field independence assumption: $Q_\phi(\mathbf{S}) = \prod_i Q_{\phi_i}(s_i)$. □

C Reproducibility Details

C.1 Prior Hyperparameters

Table 2 reports the prior hyperparameters $(\pi_i, \alpha_i, \beta_i, \sigma_i)$ used for each rule in the Togolese instantiation of RSI. These values encode institutional knowledge about historical compliance rates and rule stability.

Table 2: Prior hyperparameters by rule. π_i: activation probability. (α_i, β_i): Beta prior on compliance ($\mathbb{E}[c_i] = \alpha/(\alpha+\beta)$). σ_i: standard deviation of parametric drift prior.

Rule ID	**Description**	π_i	α_i	β_i	$\mathbb{E}[c_i]$	σ_i
R1_TVA	VAT 18%, threshold 60M / 100M FCFA	0.92	8.0	2.0	0.80	0.05
R2_IS	CIT 29%, threshold 100M FCFA	0.88	6.0	4.0	0.60	0.03
R3_IMF	Minimum flat tax 1%	0.85	9.0	1.5	0.86	0.02
R4_TPU	Informal sector flat tax	0.70	3.0	7.0	0.30	0.15

Rationale for prior choices.

- **R1_TVA**: High prior compliance ($\mathbb{E}[c_i] = 0.80$) reflects that formal VAT-registered enterprises in Togo generally file returns, even if under-declared. Low drift ($\sigma = 0.05$) reflects the stability of the 18% rate over the study period.
- **R2_IS**: Moderate compliance (0.60) reflects the complexity of CIT computation and higher avoidance rates among large enterprises.
- **R3_IMF**: High compliance (0.86) as IMF is simpler to compute (1% of turnover) and harder to avoid.
- **R4_TPU**: Low compliance (0.30) reflects the well-documented informality of the micro-enterprise sector in Sub-Saharan Africa. High drift ($\sigma = 0.15$) reflects high uncertainty about this segment.

C.2 Variational Inference Parameters

Table 3: Variational inference hyperparameters.

Parameter	Value	Rationale
Learning rate η	1.5	Amplifies posterior update beyond standard VI
Max iterations	150	Empirical convergence $\leq$ 10 iterations
Convergence tolerance ε	10^{-5}	ELBO change threshold
Decision threshold τ	0.551	Optimized for F1 on validation set

C.3 Baseline Configurations

Table 4: Baseline model configurations.

Model	Parameter	Value
XGBoost	n_estimators	150
	max_depth	4
	learning_rate	0.1
	subsample	0.8
	random_state	42
MLP	hidden_layer_sizes	(64, 32, 16)
	activation	ReLU
	max_iter	200
	early_stopping	True
	validation_fraction	0.1
	random_state	42
RBS	VAT threshold (2022–2024)	60M FCFA
	VAT threshold (2025)	100M FCFA
	Payment delay alert threshold	90 days

C.4 Computational Environment

Experiments were run on a standard desktop machine (Windows 11, Intel Core processor, 16GB RAM). All code is implemented in Python 3.11 using NumPy, SciPy, and scikit-learn. No GPU was used. Runtime for the full experiment pipeline is approximately 5–10 minutes.

C.5 Random Seeds

All experiments use `numpy.random.seed(42)` for reproducibility. Dataset generation uses `numpy.random.default_rng(42)`

References

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